INCH-POUND

MIL-PRF-39016/14E
31 July 2006
SUPERSEDING
MIL-PRF-39016/14D
20 July 1988

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, 4PDT, LOW LEVEL TO 2 AMPERES (0.150-INCH TERMINAL SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39016.

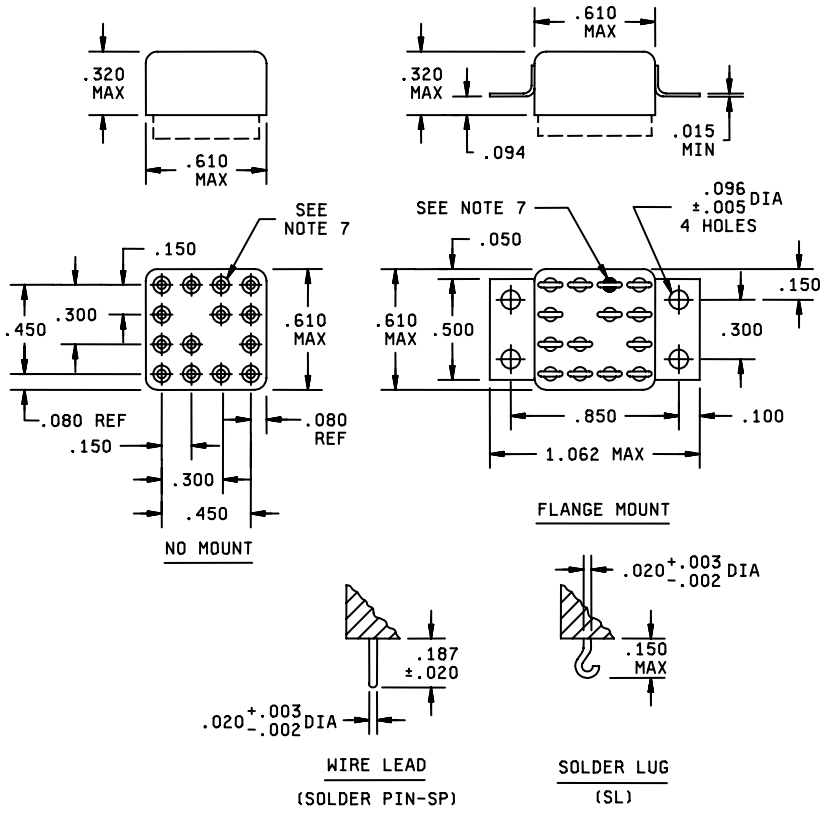
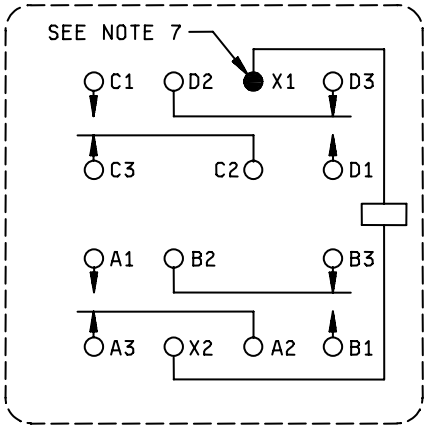


FIGURE 1. Dimensions and configuration.

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CIRCUIT DIAGRAM
TERMINAL VIEW
DEENERGIZED POSITION

Inches	mm	Inches	mm
.001	0.03	.150	3.81
.003	0.08	.187	4.75
.005	0.13	.300	7.62
.020	0.51	.320	8.13
.050	1.27	.450	11.43
.080	2.03	.500	12.70
.094	2.39	.610	15.49
.096	2.44	.850	21.59
.100	2.54	1.062	26.97

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, the tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type amounts.
5. The shape of solder lugs is optional.
6. Terminal numbers in circuit diagram are for reference only. Numbers do not appear on relay.
7. Indicated terminal shall be identified with a contrasting bead.
8. Coil symbol optional in accordance with MIL-STD-1285.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Contact data:

Load ratings:

High level (relay case grounded):

Resistive: 2 amperes at 28 V dc; 0.125 ampere at 115 V ac, (60 and 400 Hz) with case grounded. 0.5 ampere at 115 V ac, (60 and 400 Hz) with case not grounded.

Inductive: 0.5 ampere at 200 millihenries inductive at 28 V dc.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.050 ohm maximum.

High level:

During life: Not more than 5 percent of open circuit voltage.

After life: 0.150 ohm maximum.

Low level:

During life: 33 ohms maximum.

After life: 0.150 ohm maximum.

Intermediate current:

During intermediate current: 1 ohm maximum.

After intermediate current: 3 ohm maximum.

Contact bounce: 2.0 milliseconds maximum. (Applicable to failure rate level "L").

Contact stabilization time: 2.5 milliseconds maximum. (Applicable to failure rate levels "M" "P", and "R").

Overload (high level only): 4 amperes resistive at 28 V dc. 1.0 ampere inductive at 28 V dc (ac not applicable).

Coil data: See table I.

Operate time: 4.0 ms maximum over temperature range with rated coil voltage.

Release time: 4.0 ms maximum over temperature range from rated coil voltage.

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Electrical data:

Insulation resistance: 10,000 megohms minimum, except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage:

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and between all contacts in the energized and deenergized positions	750	
Between case, frame, or enclosure and coil(s)	500	350
Between all contacts and coil(s)	750	All terminals to case
Between open contacts in the energized and deenergized positions	500	
Between contact poles	750	

Environmental data:

Temperature range: -65°C to +125°C.

Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Vibration (random): MIL-STD-202, method 214, test condition IG. Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts. Applicable to qualification and group C testing only.

Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts, and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Physical:

Terminals: See figure 1 and table I.

Terminal strength: 1.5 ±0.2 lbs. (pull).

Solderability: Applicable.

Terminal twist test: Applicable to wire lead terminals.

Dimensions and configuration: See figure 1.

Weight: 8.5 grams (0.30 ounce) maximum.

Identification marking (full): Applicable.

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Life test requirements:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or Identifying Number (PIN): M39016/14-(dash number from table I and suffix letter designating failure rate level).

TABLE I. Dash numbers and characteristics. 1/

Dash number 2/	Mount	Leads	Coil voltage V dc 3/		At 25°C				Over temperature range		
			Rated	Max	Coil resistance ohms ±10%	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)	Specified pickup value (voltage) (V dc)	Specified hold value (voltage) (V dc)	Specified dropout value (voltage) (V dc)
004	Flange	Lug	6	7	28	2.7	1.6	0.3	3.8	2.2	0.18
005	No mount	Wire (SP)									
009	Flange	Lug	9	11	73	4.2	2.5	0.4	5.6	3.4	0.25
010	No mount	Wire (SP)									
006	Flange	Lug	12	14	115	5.4	3.2	0.6	7.6	4.3	0.36
007	No mount	Wire (SP)									
011	Flange	Lug	18	22	280	8.4	5.0	0.8	11.2	6.7	0.5
012	No mount	Wire (SP)									
013	Flange	Lug	22	27	430	10.3	6.0	1.0	14.0	8.0	0.6
003	No mount	Wire (SP)									
001	Flange	Lug	26.5	35	720	13.5	8.1	1.5	18.0	10.8	0.9
008	Flange	Wire (SP)									
002	No mount	Wire (SP)									
014	Flange	Lug	36	42	1,040	17.1	10.5	1.9	22.8	14.0	1.1
015	No mount	Wire (SP)									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuit are not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 operations): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example, 001L - - - - - 003R.

3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.

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TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.	M39016/14-001	18 units plus 1 open unit for level L at C = 0 2/ 33 units plus 1 open unit for level M at C = 0 2/ Qualification inspection as applicable.
	M39016/14-004 M39016/14-006 M39016/14-009 M39016/14-011 M39016/14-014	2 units, each part number, qualification inspection table Q1.
	M39016/14-003	2 units, qualification inspection table, Q1, and shock, vibration, acceleration, terminal strength, and seal.

1/ For retention of qualification or extension of qualification to low failure rate, all life test data accumulated on MIL-PRF-39016/53 and MIL-PRF-39016/54 may be used in addition to MIL-PRF-39016/14 data. Prior to retention of qualification testing, the relay manufacturer shall preselect the sampling plans.

2/ The number of units required for qualification testing will be increased as required in group IV, table II, MIL-PRF-39016, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to qualification testing, the relay manufacturer shall preselect the sampling plan.

Qualification inspection (reduced testing) and sample size: See table III. If the relays produced for MIL-PRF-39016/14 are similar in construction and design except for the diodes and coils to the relays produced for MIL-PRF-39016/53 and MIL-PRF-39016/54, then reduced testing for qualification of MIL-PRF-39016/14 relays may be performed concurrent with or subsequent to successful qualification of MIL-PRF-39016/53 and MIL-PRF-39106/54.

TABLE III. Qualification inspection (reduced testing).

Examination or test
2 units each coil voltage - Q1 of qualification inspection table. 1 unsealed sample unit for internal examination.

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Supersession data: See table IV.

TABLE IV. Supersession data.

New PIN M39016/14 1/	Superseded PIN M5757/80-
004	016
006	017
001	018
005	025
007	026
002	027

1/ Complete PIN shall contain suffix letter L, M, P, or R to designate failure rate level (see 2/ of table I). A part with any failure rate supersedes the applicable MIL-R-5757 part.

Referenced documents: In addition to MIL-PRF-39016, this document references the following:

MIL-R-5757 MIL-PRF-39016/53 MIL-PRF-39016/54
MIL-STD-202 MIL-STD-1285

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5945-1282-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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